

Office Network Design using VLANs and Router-on-a-Stick

Project Description (English):

I designed a professional office network composed of three departments (HR, Finance, IT), each connected through separate VLANs and integrated using a Router-on-a-Stick setup. All devices were assigned IP addresses with appropriate gateways. Connectivity between departments was verified using ping tests. The project demonstrates practical knowledge of VLAN segmentation, IP planning, and inter-VLAN routing.

Tools Used:

- Cisco Packet Tracer
- 3 Switches (2960)
- 1 Router (ISR or 2911)
- PCs for each department
- Straight-through cables

Project Steps:

1. Created 3 VLANs: HR (10), Finance (20), IT (30).
2. Assigned access ports on each switch to the correct VLAN.
3. Connected switches to router via trunk ports (g0/0, g0/1, g0/2).
4. Configured router sub-interfaces with encapsulation dot1Q and IP addresses.
5. Assigned IP addresses and gateways to each PC manually.
6. Verified connectivity between devices using ping.